

Artifact 10 — Agentic Safety Discovery

Risk & Ethics Gating · Stage 4

DOCUMENT DETAILS

Project	Home-Care-AI	Stage	Discovery → Stage 4 (Risk & Ethics Gating)
Date	2026-03-26	Updated	2026-03-27 — Three new findings backported from Artifact 16 (Compliance Privacy Audit): G-CC-1 (§5A), G-DS-05 (§5C), SC-07 (§9)
Methodology	Level 1 (Informer) / Level 2 (Verifier) / Level 3 (Escalator); Fallback Protocol + False Positive cost for every Level 3 action		
Input	Artifact 9 — Ethics & Trust Map (notification routing rules §5, feature clearance §7, compound combinations §4); CLAUDE.md Article VIII Agentic Control Matrix		
Regulatory	Australian Privacy Act 1988 (APP). HIPAA-grade security applied as design floor.		
Feeds into	Artifact 11 — Brainstorm Experiments (Skill 11); agentic-logic-spec (Execution Plugin)		

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1. Autonomy Level Definitions (CLAUDE.md Article VIII)

Level	Name	Definition	HITL Required	Reversibility
Level 1	Informer	Passive observation, computation, or notification. No external commitment created.	No	Fully reversible (display only)
Level 2	Verifier	Generates a recommendation or draft action. Requires human confirmation before the action executes. Borderline signals or moderate-consequence actions.	Yes — gates execution	Reversible before confirmation
Level 3	Escalator	Executes an irreversible or high-consequence external action. Creates a real-world obligation (message sent, assignment committed, record modified).	Mandatory HITL approval	Irreversible after execution

2. Agentic Control Matrix — Full Action Inventory

All actions the system can autonomously take, classified by autonomy level. Sources: Artifact 5 (OST solutions), Artifact 6 (Top 5 features), Artifact 9 (notification routing rules).

2A — Vacancy Detection & Matching

Action ID	Action	Level	Rationale	HITL Gate
ACT-V-01	Detect unconfirmed visit slot (shift cancellation received or visit not started within window)	■ L1 Informer	Pure event detection; no external action taken; purely observational	None
ACT-V-02	Run SPP-weighted match algorithm against available carers	■ L1 Informer	Computation only; result not surfaced externally; coordinator is sole recipient	None
ACT-V-03	Compute proximity score per candidate (suburb postcode → client suburb)	■ L1 Informer	Derived from self-reported postcode (S-3); no GPS; Green zone data only	None
ACT-V-04	Rank candidates by composite SPP match score	■ L1 Informer	Internal ranking; no external communication; reversible at any point	None
ACT-V-05	Push vacancy alert to coordinator ("Vacancy detected — review now")	■ L1 Informer	Notification only; no action committed; coordinator has full control	None
ACT-V-06	Display ranked shortlist in coordinator UI with match scores	■ L1 Informer	Display only; coordinator not yet committed to any candidate	None
ACT-V-07	Surface match explanation to coordinator (which SPP tags drove score)	■ L1 Informer	Coordinator-facing only. CC-6 constraint applies: match explanation (A-2) must never appear in carer notification or family notification payload. Internal display only.	None

2B — Coordinator Approval Flow (HITL Gate)

Action ID	Action	Level	Rationale	HITL Gate
ACT-A-01	Present 3-Tap Approval interface to coordinator	■ L1 Informer	Interface presentation only; no action until tap 3	None
ACT-A-02	Request coordinator to confirm replacement carer selection	■ L2 Verifier	Coordinator decision required before any external action. This is the primary HITL gate for the entire downstream action chain.	Coordinator must tap "Confirm" — system waits
ACT-A-03	Record coordinator approval decision (coordinator_id + timestamp + candidate_id)	■ L1 Informer	Immutable audit record of human decision; no external action	None
ACT-A-04	Coordinator overrides AI ranking (selects non-top candidate)	■ L1 Informer	Override is a coordinator decision; system records it; no resistance or friction permitted	None

2C — Carer Assignment (Level 3 — Coordinator Approval Required)

Action ID	Action	Level	Rationale	HITL Gate	Fallback	False Positive Cost
ACT-C-01	Send WhatsApp assignment message to confirmed replacement carer	■ L3 Es calator	External, irreversible message. Carer receives a job offer and may accept or decline. Creates real-world scheduling commitment. Cannot be unsent. CC-8 applies: Green data only (carer first name, client suburb, visit time, match score).	coordinator_approved = true AND candidate_id confirmed in ACT-A-02	If coordinator does not approve within SLA → escalate to backup coordinator (see §3 HITL Timeout). If no approval → no message sent. Vacancy remains open.	High. Wrong carer assigned → unfamiliar person arrives at vulnerable senior's home. Client distress, potential refusal of entry (P-6 risk). Carer's day disrupted unnecessarily.
ACT-C-02	Send carer pre-visit briefing notification	■ L3 Es calator	External, irreversible message. Briefing reveals structured SPP fields (entry protocol, personal sensitivities — P-5 phrased as operational guidance only). CC-6 applies: match explanation (A-2) must NOT appear. P-2 (gender preference) must NOT appear.	ACT-C-01 complete (carer assignment confirmed)	If ACT-C-01 failed, briefing is not sent.	High. Wrong briefing → carer arrives unprepared. Correct briefing sent to wrong carer → SPP data disclosed to unauthorised recipient.

2D — Client Notification (Level 3 — E-3 Gate)

Action ID	Action	Level	Rationale	HITL Gate	Fallback	False Positive Cost
ACT-P-01	Send client notification ("Your visit today will be with [first name]")	■ L3 Es calator	External, irreversible notification to vulnerable senior. Incorrect notification → client confusion and anxiety, especially for familiarity-threshold clients (P-3). First in E-3 notification chain — must precede family notification.	coordinator_approved = true AND ACT-C-01 complete	If client notification channel unavailable (no app/SMS enrolment), log CLIENT_NOTIFICATION_UNAVAILABLE and proceed — client_notified flag set to false. Family notification must NOT proceed.	High. Wrong carer's name → client refuses entry or is distressed by unexpected visitor. Missed notification → client has no warning, increases risk for dementia clients (P-3/P-4).
ACT-P-02	Update client's SPP continuity history with new visit record	■ L2 Verifier	Modifies a persistent client record (P-7 + P-8 binary). Yellow zone data. Coordinator review required if it contradicts existing record.	ACT-C-01 complete AND visit completed (post-visit trigger)	If visit did not occur (carer cancelled, client refused), SPP update is suppressed. No counterfactual added.	Medium. Incorrect continuity record → distorted familiarity scores in future matching. Adds noise but not immediately harmful.

2E — Family Notification (Level 3 — E-3 Double Gate)

Action ID	Action	Level	Rationale	HITL Gate	Fallback	False Positive Cost
ACT-F-01	Send family notification ("Replacement carer arranged for [client first name]'s visit")	■ L3 Escalator	External, irreversible message. E-3 structural gate: family is never first. Violating this is the Arthur Kovacs failure case — family in panic before client knows. CC-8 applies if WhatsApp : Green data only.	coordinator_approved = true AND client_notified = true (or client_notification_preference = family_first with recorded explicit consent)	If client_notified = false and no family-first preference recorded: family notification is suppressed . Log FAMILY_NOTIFICATION_SUPPRESSED_E3_GATE. Alert coordinator.	Critical (trust-destroying) . Family notified before client → family calls client in panic. Client feels surveilled, not cared for. This is the exact Arthur Kovacs complaint (SR-002). Destroys trust in the product with the entire family unit — a multi-stakeholder loss.

2F — Compliance Monitoring

Action ID	Action	Level	Rationale	HITL Gate
ACT-CM-01	Flag overdue care plan review (A-3)	■ L1 Informer	Coordinator-facing only; display alert; no external action	None
ACT-CM-02	Flag expired carer credential (A-3)	■ L1 Informer	Coordinator-facing only; display alert; prevents expired carer appearing in match candidates	None
ACT-CM-03	Suppress expired-credential carer from shortlist	■ L1 Informer	Automatic filter — not a punitive action; carer is not notified; coordinator is informed	None
ACT-CM-04	Escalate unresolved compliance item to agency owner	■ L2 Verifier	Higher-consequence action (escalates above coordinator); requires coordinator inaction for defined SLA before triggering	compliance_sla_expired = true (coordinator did not act within window)
ACT-CM-05	Lock coordinator from approving an assignment involving an expired-credential carer	■ L2 Verifier	UI-level enforcement gate; coordinator cannot bypass without explicit override acknowledgement; override is logged	Coordinator must acknowledge override with reason

2G — SPP Management

Action ID	Action	Level	Rationale	HITL Gate
ACT-S-01	Pre-populate familiarity history from past visit records	■ L1 Informer	Computation from Green data (P-7 visit log); coordinator reviews before SPP is published	None (but coordinator review recommended before activation)
ACT-S-02	Suggest SPP field update to coordinator (new pattern observed)	■ L1 Informer	Suggestion only; no write occurs; coordinator decides	None
ACT-S-03	Write coordinator-confirmed SPP update to client record	■ L2 Verifier	Persistent modification of Yellow-zone client record; requires coordinator explicit confirmation	Coordinator must confirm each field update
ACT-S-04	Calculate SPP completeness score (P-10)	■ L1 Informer	Aggregate metadata; no individual field content; operational dashboard only	None

2H — System & Audit

Action ID	Action	Level	Rationale	HITL Gate
ACT-AU D-01	Write immutable audit log entry (APPAuditLogEntry)	■ L1 Informer	Mandatory for every state transition; append-only; no HITL required; automated by design	None — this action must NEVER be gateable or defeatable
ACT-AU D-02	Generate audit report for agency owner / regulatory review	■ L1 Informer	Read-only; no modification; access-controlled	None
ACT-AU D-03	Mark incident as RESOLVED	■ L2 Verifier	Closes the state machine for a vacancy incident; should not self-close without coordinator confirmation	Coordinator confirmation required

3. HITL Timeout Protocol — Vacancy Matching Context

Adapted from CLAUDE.md Article VIII for the home care scheduling context. Unlike clinical emergency dispatch, the vacancy workflow is time-critical (visit is hours away) but not life-critical. Timeouts trigger escalation, not auto-dispatch.

```
ACT-A-02: Coordinator approval requested
■
■■■ Coordinator responds within SLA (30 min)
■ → Confirm or Override → ACT-C-01 executes → continue flow
■
■■■ Coordinator timeout (30 min)
■
  → Log HITL_TIMEOUT (coordinator tier)
  → Push escalation alert to backup coordinator / agency owner
    ■
    ■■■ Backup responds within SLA (15 min)
    ■ → Confirm → ACT-C-01 executes
    ■
    ■■■ Backup timeout (15 min)
    ■
      → Log HITL_DOUBLE_TIMEOUT
      → Log VACANCY_UNRESOLVED
      → Alert agency owner: "Vacancy unresolved – manual action required"
      → System takes NO autonomous action
      [No auto-dispatch of carer without human approval – safety default]
```

Critical distinction from clinical emergency protocol: In the clinical domain, HITL double-timeout → auto-dispatch (EMS) because inaction costs a life. In the matching domain, HITL double-timeout → no action + escalation alert, because auto-assigning a carer to a vulnerable senior without any human approval is a higher risk than a missed visit. The agency must resolve the vacancy through manual means.

HITL SLA Register

Action	Tier 1 HITL	SLA	Timeout Consequence	Tier 2 HITL	SLA	Double Timeout Consequence
ACT-A-02 (coordinator approval)	Coordinator	30 min	HITL_TIMEOUT + escalate	Backup coordinator / agency owner	15 min	HITL_DOUBLE_TIMEOUT + VACANCY_UNRESOLVED — no autonomous action
ACT-CM-04 (compliance escalation)	Coordinator	Defined per agency SLA (default 48h)	COMPLIANCE_SLA_EXPIRED flag	Agency owner	24h	COMPLIANCE_CRITICAL_OPEN — external audit flag
ACT-P-02 (SPP update post-visit)	Coordinator	24h post-visit	Suppressed (no write) + flag for next session	None	—	SPP record remains unchanged

4. Fallback Protocol & False Positive Cost — All Level 3 Actions

CLAUDE.md Article III Stage 4 requirement: "Fallback Protocol + False Positive cost for every Level 3 action."

ACT-C-01 — WhatsApp Assignment to Carer

	Detail
Fallback Protocol	If coordinator HITL times out at both tiers → no message sent → log VACANCY_UNRESOLVED → human manages manually. If WhatsApp delivery fails (CC-8 channel unavailable) → fallback to SMS → if SMS fails → push alert to coordinator: "Carer notification failed — call carer directly." System never silently drops the assignment notification.
False Positive (wrong carer assigned)	Carer receives unexpected assignment. If they accept: wrong person arrives at vulnerable client's home. Depending on P-3 (familiarity threshold) this may trigger client refusal of entry. Carer's day disrupted. Continuity record incorrectly updated. Cost: High.
False Negative (no carer assigned)	Visit remains unfilled. Client has no carer today. Family is not notified (E-3 gate). Coordinator must resolve manually. For a high-familiarity-threshold client (P-3 = "known carers only"), no familiar carer = no visit. Cost: High — client safety risk.
Recovery Action	ACT-C-01 reversal is not possible (message sent). Recovery is: coordinator calls carer directly to cancel if wrong assignment; log ASSIGNMENT_CANCELLED with coordinator_id and reason; do NOT auto-send a correction message.

ACT-C-02 — Carer Pre-Visit Briefing Notification

	Detail
Fallback Protocol	If ACT-C-01 failed → briefing is not sent (no orphaned briefing without assignment). If briefing delivery fails → coordinator is alerted: "Briefing not delivered — brief carer verbally." System logs BRIEFING_DELIVERY_FAILED.
False Positive (wrong briefing sent)	SPP data (P-5 personal sensitivities, entry protocol) disclosed to wrong carer. This is an APP disclosure breach. Cost: Critical. Coordinator must be alerted immediately. Wrong recipient must be logged. No corrective re-briefing sent without coordinator review.
False Negative (briefing not delivered)	Carer arrives unprepared. For high-P-3 or P-4-flagged clients, unprepared carer may cause distress or client refusal. Cost: High.
CC-6 Guard (mandatory):	Before any briefing sends, system must assert: match_explanation_in_payload = false AND gender_preference_in_payload = false. If either assertion fails → briefing is blocked → coordinator alerted → manual briefing required.

ACT-P-01 — Client Notification

	Detail
Fallback Protocol	If client has no app/SMS enrolment: log CLIENT_NOTIFICATION_UNAVAILABLE, set client_notified = false. This suppresses ACT-F-01 (family notification). Coordinator is alerted to notify client manually before family notification can proceed.
False Positive (wrong carer name in notification)	Client told a carer name that doesn't match the person who arrives. For familiarity-threshold clients (P-3), this is a trigger for distress and possible refusal. Cost: High — client harm risk.
False Negative (notification not sent)	Notification channel unavailable → client_notified = false → family notification gate blocked (E-3 compliant) → manual follow-up required. Cost is coordinator time. Cost: Medium.
E-3 Guard (mandatory):	Client notification is always attempted before family notification is permitted. Suppression of client notification does not unlock family notification — it locks it.

ACT-F-01 — Family Notification

	Detail
Fallback Protocol	If E-3 gate fails (client_notified = false and no family-first consent): notification is suppressed entirely . Log FAMILY_NOTIFICATION_SUPPRESSED_E3_GATE. Coordinator alerted. Family remains uninformed until coordinator decides to notify manually. This is intentional — the system must not be the first to panic the family.
False Positive (family notified before client)	The Arthur Kovacs failure case. Family calls client in a panic before client knows there has been a scheduling change. Client feels monitored, not supported. Trust destroyed with both client and family. Cost: Critical — existential trust loss.
False Positive (incorrect replacement name in family notification)	Family given wrong carer name → family follows up with agency → confusion → trust erosion. Cost: High.
False Negative (family not notified after client is informed)	Family remains unaware of replacement. Lower risk than premature notification. Family may call agency if concerned. Cost: Low — family can self-serve.
Recovery Action	If wrong information was sent → coordinator must contact family directly. System does NOT send a corrective automated message (second message compounds confusion). Log FAMILY_NOTIFICATION_ERROR with coordinator_id.

5. Ethics-Derived Safety Rules — Agentic Guardrails from Artifact 9

Compound combination flags and notification routing rules from Artifact 9 translated into mandatory IF-THEN guards. These become structural constraints in agentic-logic-spec pseudocode.

5A — Compound Combination Guards

Guard ID	Compound	IF	THEN	Severity if Violated
G-CC-4	S-4 + P-2 (carer history + gender pref)	Match algorithm uses both S-4 (carer assignment history) AND P-2 (gender preference) as active scoring inputs	Assert: legal opinion on E-1 (anti-discrimination) is signed_off = true. If signed_off = false → P-2 is removed from scoring, surfaced to coordinator as advisory only. S-4 may remain as scoring input.	CRITICAL — blocks matching engine build
G-CC-6	A-2 + carer notification	Before ACT-C-02 (briefing) executes	Assert: match_explanation_in_payload = false AND gender_preference_in_payload = false. If either = true → block send → alert coordinator	HIGH — APP disclosure breach
G-CC-8	WhatsApp + Yellow/Red data	Before ACT-C-01 executes (WhatsApp send)	Assert: all payload fields are Green zone only. Enumerate allowed fields: carer_first_name (S-1), visit_time (Green), client_suburb (not full address), match_score (A-1 numeric). If any Yellow/Red field detected → strip field → log FIELD_STRIPPED_CC8. If stripping makes message incomplete → block send → alert coordinator	HIGH — APP 8 cross-border disclosure breach
G-CC-5	Free-text + any SPP field (P-9)	P-9 (free-text notes) is present in v1	Block collection. P-9 is deferred to v2. If field exists in schema → log P9_COLLECTION_BLOCKED. No free-text field appears in any notification payload.	CRITICAL — uncontrolled PHI risk
G-CC-3	A-4 + P-11 in external surface	Continuity score + client address appear together in any family-facing or client-facing output	Block. Continuity score is coordinator-only. Client address (full) is coordinator + confirmed-carer only. Assert: these two fields never appear in the same external response payload.	HIGH — health inference + location inference combined
G-CC-1 (added Artifact 16)	P-3 + P-4 + P-5 (familiarity threshold + cognitive flags + personal sensitivities)	All three fields present together in any external payload	Block send. These three fields combined create a near-complete psychological and cognitive vulnerability profile of a senior (effectively a dementia inference without a medical record). DPIA mandatory before collection of P-3 + P-4 together is enabled — not just before build. Assert: NOT (P3_present AND P4_present AND P5_present IN payload). Log CC1_COMPOUND_BLOCKED. Alert coordinator.	CRITICAL — health inference compound; DPIA required

5B — Notification Order Guards (E-3 Enforcement)

Guard ID	Rule	IF	THEN
G-E3-01	Family never first	family_notification_triggered = true	Assert: coordinator_approved = true AND (client_notified = true OR client_notification_preference = family_first with recorded_consent = true). If assertion fails → block family notification → log E3_GATE_BLOCKED
G-E3-02	Client notification failure locks family gate	client_notified = false (channel unavailable)	Family notification remains suppressed. Coordinator is alerted. Gate cannot be bypassed by system.

Guard ID	Rule	IF	THEN
G-E3-03	AI inference to family — blocked	Any AI-generated health signal (A-3, A-4, any inferred value)	Must pass: coordinator review → clinical framing → client disclosure → THEN family (if client consents). AI inference is never routed directly to family.
G-E3-04	Compliance alerts — coordinator only	Any compliance alert (A-3)	Must not appear in client-facing or family-facing UI. Coordinator only. System asserts <code>recipient_role = COORDINATOR</code> before surfacing.

5C — Data Sensitivity Guards

Guard ID	Rule	Trigger	Action
G-DS-01	No PHI in notification payloads	Any outbound message construction	Strip: no full name + condition, no P-4 flag content, no P-6 refusal reason. Allowed: first name only (if Green), visit time, suburb, entry protocol (label only — not reason).
G-DS-02	P-11 — suburb until confirmed	Client address in pre-assignment matching	Before ACT-A-02 completes: client suburb only. After ACT-A-02 (<code>coordinator_approved = true</code>): full address released to confirmed carer only. Never in WhatsApp.
G-DS-03	P-8 — binary only in v1	Carer familiarity history update	Write: <code>carer_uuid + visit_count + last_visit_date</code> . Do NOT write <code>outcome_rating</code> . Assert: <code>outcome_rating</code> field absent from v1 schema.
G-DS-04	A-2 — coordinator UI only	Match explanation	Assert: A-2 field scoped to <code>coordinator_session</code> only. Never included in: carer notification, client notification, family notification, audit log display to external parties.
G-DS-05 (added Artifact 16)	Client notification phrasing — P-3-aware branching	ACT-P-01 constructs client notification message	Notification phrasing must branch on <code>client.familiarity_threshold</code> value. For $P-3 \geq 2$ ("known carers only" / cognitively impaired): do NOT use "who has visited you before" — client may not remember; this causes confusion, not reassurance. Use: "Your carer today is [first name], arranged by Angela." For $P-3 < 2$: standard phrasing ("who has visited you before") is appropriate. Rationale: "who has visited you before" implies the client will remember — an unsafe assumption for high-P-3 clients.

6. Edge Cases — Mandatory Explicit Handling

Edge Case	Scenario	Required Handling
EC-01	Coordinator approves; carer declines WhatsApp assignment	ACT-C-01 sent → carer responds "no" (or no response within 15 min). Log CARER_DECLINED. Coordinator re-enters matching flow with next candidate. State returns to HITL_PENDING (not RESOLVED).
EC-02	No eligible candidate in shortlist (all fail SPP gate)	Match returns 0 candidates. Alert coordinator: "No SPP-eligible carers available — consider broadening criteria." Do not auto-select outside SPP. Coordinator must explicitly override and acknowledge the SPP mismatch.
EC-03	Client P-3 = "Known carers only" and no familiar carer available	All familiar carers unavailable AND override is unfamiliar carer. Coordinator must acknowledge: "Assigning an unfamiliar carer to a familiarity-threshold client." This is L2 Verifier — requires explicit coordinator override acknowledgement. Log FAMILIARITY_THRESHOLD_OVERRIDE with coordinator_id.
EC-04	G-CC-4 legal sign-off not present (E-1 unresolved)	P-2 (gender preference) is collected but E-1 legal sign-off is false. P-2 must not enter match scoring. System shows P-2 to coordinator as "Client preference advisory (not scored)." SPP field remains stored; algorithm ignores it.
EC-05	WhatsApp message delivery failure	CC-8 channel fails. Fallback: attempt SMS. If SMS fails: alert coordinator "Carer notification failed — call [carer name] directly." Never drop silently. Log CARER_NOTIFICATION_FAILED + channel attempted.
EC-06	Duplicate vacancy event (system detects the same slot twice)	Idempotency check: if incident_id already exists in open state, suppress duplicate. Log DUPLICATE_VACANCY_SUPPRESSED. Do NOT trigger a second coordinator notification for the same incident.
EC-07	Coordinator selects a carer with an expired credential	ACT-CM-05 fires: UI gate prevents confirmation. Coordinator must acknowledge override. Override logged with coordinator_id + reason. Expired-credential assignment is a compliance risk — log as COMPLIANCE_OVERRIDE_ACKNOWLEDGED.
EC-08	Client has client_notification_preference = family_first but no recorded consent	Family-first preference exists in UI without consent timestamp. Do not use. E-3 gate requires recorded_consent = true. Alert coordinator: "Family-first preference requires a recorded consent event — update client record." Default to standard E-3 order.
EC-09	SPP update conflicts with existing record (coordinator updates, AI suggests differently)	Human always wins. AI suggestion is suppressed in favour of coordinator's explicit entry. Log AI_SUGGESTION_OVERRIDDEN + coordinator_id. Never silently blend AI suggestion with coordinator entry.
EC-10	HITL double-timeout (both coordinator tiers unresponsive)	Log HITL_DOUBLE_TIMEOUT. Log VACANCY_UNRESOLVED. Alert agency owner via push notification and email. System makes NO autonomous carer assignment. Vacancy remains open. Agency resolves manually.

7. State Machine — Vacant Visit Matching Flow

Mermaid diagram: three paths — Happy Path, HITL Timeout, and E-3 Gate Failure. Mandatory per CLAUDE.md Article V Stage B (agentic-logic-spec requirement preview).

```
stateDiagram-v2
    [*] --> NORMAL

    NORMAL --> VACANCY_DETECTED : Shift cancellation received\nor visit window missed
    VACANCY_DETECTED --> MATCHING : ACT-V-01 to ACT-V-04\n(L1 – no external action)
    MATCHING --> SHORTLIST_READY : Candidates ranked\n(L1 – coordinator display only)
    SHORTLIST_READY --> HITL_PENDING : ACT-A-02\nCoordinator approval requested\n(L2 – HITL gate)

    %% Happy Path
    HITL_PENDING --> COORDINATOR_APPROVED : Coordinator confirms\nwithin 30-min SLA
    COORDINATOR_APPROVED --> CARER_NOTIFIED : ACT-C-01\nWhatsApp sent\n(L3 – CC-8 guard passes)
    CARER_NOTIFIED --> BRIEFING_SENT : ACT-C-02\nBriefing sent\n(L3 – CC-6 guard passes)
    BRIEFING_SENT --> CLIENT_NOTIFIED : ACT-P-01\nClient SMS/app\n(L3 – E-3 gate 1)
    CLIENT_NOTIFIED --> FAMILY_NOTIFIED : ACT-F-01\nFamily notification\n(L3 – E-3 gate 2:\nclient_notified
= true)
    FAMILY_NOTIFIED --> RESOLVED : Visit completed\nACT-AUD-01: INCIDENT_RESOLVED

    %% HITL Timeout Path
    HITL_PENDING --> HITL_TIMEOUT_T1 : 30 min – no coordinator response
    HITL_TIMEOUT_T1 --> HITL_PENDING_T2 : Escalate to backup coordinator
    HITL_PENDING_T2 --> COORDINATOR_APPROVED : Backup confirms
    HITL_PENDING_T2 --> HITL_DOUBLE_TIMEOUT : 15 min – no backup response
    HITL_DOUBLE_TIMEOUT --> VACANCY_UNRESOLVED : No autonomous action\nAgency owner alerted
    VACANCY_UNRESOLVED --> [*] : Manual resolution required

    %% E-3 Gate Failure Path
    CLIENT_NOTIFIED --> FAMILY_GATE_BLOCKED : client_notified = false\nE-3 gate blocks family
    FAMILY_GATE_BLOCKED --> COORDINATOR_ALERTED : Coordinator must\nnotify client manually
    COORDINATOR_ALERTED --> CLIENT_NOTIFIED : Manual client notification\ncompleted

    %% CC-8 Guard Failure
    CARER_NOTIFIED --> CARER_NOTIFICATION_FAILED : CC-8 guard blocks\nor delivery failure
    CARER_NOTIFICATION_FAILED --> COORDINATOR_ALERTED : Coordinator must\ncall carer directly
```

8. Autonomy Classification Summary

Level	Action Count	Actions
■ L1 Informer	16	ACT-V-01 through ACT-V-07, ACT-A-01, ACT-A-03, ACT-A-04, ACT-CM-01, ACT-CM-02, ACT-CM-03, ACT-S-01, ACT-S-02, ACT-S-04, ACT-AUD-01, ACT-AUD-02
■ L2 Verifier	6	ACT-A-02 (primary HITL gate), ACT-P-02 (SPP update), ACT-CM-04 (compliance escalation), ACT-CM-05 (credential lock), ACT-S-03 (SPP write), ACT-AUD-03 (incident close)
■ L3 Escalator	4	ACT-C-01 (WhatsApp carer assignment), ACT-C-02 (carer briefing), ACT-P-01 (client notification), ACT-F-01 (family notification)

Safety principle: The product takes 16 actions autonomously, requires human confirmation for 6 actions, and treats 4 external communications as irreversible escalations requiring prior HITL approval. The ratio is intentional — computation is autonomous, external commitment always requires a human.

9. Critical Path — Safety Constraints to Resolve Before Build

ID	Constraint	Gate Blocks	Status
SC-01	E-1 legal sign-off (anti-discrimination opinion on P-2 in matching scoring)	G-CC-4 — PM-1 Smart Match Engine build	Open — blocks matching engine
SC-02	DPIA-07 + APP 8 privacy counsel sign-off for WhatsApp cross-border disclosure	G-CC-8 — ENG-2 production	Open — blocks WhatsApp integration
SC-03	E-3 structural gate implemented in code as <code>coordinator_approved AND client_notified</code> prerequisite for family notification	ACT-F-01 — family notification	Open — must be in agentic-logic-spec
SC-04	G-CC-6 briefing guard implemented as assertion before ACT-C-02	ACT-C-02 — carer briefing	Open — must be in agentic-logic-spec
SC-05	P-9 (free-text) excluded from v1 schema entirely	G-CC-5 — all data collection	Open — must be confirmed in data model spec
SC-06	HITL double-timeout must never auto-assign a carer	EC-10 — timeout fallback	Open — must be in agentic-logic-spec
SC-07 (added Artifact 16)	APP 8 compliance review for Google Maps API proximity computation (ACT-V-03). Only carer postcode is transmitted — no client PII. However, Google's servers are US-based; cross-border disclosure confirmation required. Likely resolvable via Google Cloud DPA review — lower risk than WhatsApp but must be confirmed before ACT-V-03 ships.	ACT-V-03 — proximity score computation	Open — legal confirmation required; lower priority than SC-02

10. Handshake Table

ID	From Artifact	To Next Skill	What Transfers
HS-DISC-SAF-01	Artifact 10 — Full Agentic Control Matrix (§2)	agentic-logic-spec (Execution Plugin)	Every L1/L2/L3 classification + HITL gate condition becomes a named constant and state gate in the logic spec pseudocode
HS-DISC-SAF-02	Artifact 10 — Compound combination guards G-CC-1 through G-CC-8 (§5A) + data sensitivity guards G-DS-01 through G-DS-05 (§5C)	agentic-logic-spec (Execution Plugin)	Each guard becomes an explicit IF-THEN assertion in pseudocode, labelled with guard ID for traceability to audit log. G-CC-1 (P-3+P-4+P-5 compound) and G-DS-05 (P-3-aware notification phrasing) added 2026-03-27 from Artifact 16.
HS-DISC-SAF-03	Artifact 10 — E-3 notification order guards G-E3-01 through G-E3-04 (§5B)	agentic-logic-spec (Execution Plugin)	coordinator_approved AND client_notified prerequisite is a named boolean gate in every notification function signature
HS-DISC-SAF-04	Artifact 10 — HITL SLA register (§3)	agentic-logic-spec (Execution Plugin)	SLA values (30 min T1, 15 min T2) become THRESHOLD constants; HITL_DOUBLE_TIMEOUT state becomes an explicit node in all Mermaid diagrams
HS-DISC-SAF-05	Artifact 10 — Fallback Protocols §4 (all L3 actions)	user-stories (Execution Plugin)	Each fallback becomes an acceptance criterion scenario: "Given [condition], When [failure], Then [fallback action]" — linked to APPAuditLogEntry fields
HS-DISC-SAF-06	Artifact 10 — Edge cases EC-01 through EC-10 (§6)	brainstorm-experiments-new (Artifact 11)	EC-02 (no eligible candidate), EC-03 (familiarity override), EC-04 (E-1 P-2 advisory), EC-10 (double timeout) are the highest-priority experiment targets
HS-DISC-04	Artifact 10 — Agentic Control Matrix	agentic-logic-spec (Execution Plugin — HS-DISC-04 per master map)	Full L1/L2/L3 classification + state machine as the primary input to the logic spec

11. Action Items

Date	Owner	Action
2026-03-27	PM Lead	Share this artifact with Privacy Counsel — SC-02 (APP 8 / WhatsApp) is a production blocker for ENG-2
2026-03-27	Legal Counsel	SC-01: Anti-discrimination opinion on P-2 in match scoring — required before PM-1 matching engine build begins
2026-03-28	PM Lead	Confirm SC-03 (E-3 gate) and SC-06 (no auto-assign on timeout) are captured in agentic-logic-spec brief — these must appear as explicit pseudocode gates, not as comments
2026-03-28	PM Lead	Run brainstorm-experiments-new (Skill 11) — feed EC-02, EC-03, EC-04, EC-10 as highest-priority experiment targets
2026-03-28	Designer	Confirm that 3-Tap Approval Flow (DES-1) UI handles EC-02 (empty shortlist) and EC-03 (familiarity override) — both require explicit coordinator acknowledgement screens
2026-03-29	Engineer	Confirm that ACT-AUD-01 is implemented as a non-bypassable append to immutable store — no application path may skip the audit log write, including timeout and failure paths
2026-04-01	Legal	SC-07: Review Google Cloud / Maps API DPA against APP 8 requirements. Confirm only carer postcode is transmitted; no client PII. Likely a documentation task, not a blocker — but must be confirmed before ACT-V-03 ships to production.
2026-04-12	Privacy Officer	G-CC-1: DPIA required before collection of P-3 + P-4 combination is enabled. Scope: data flow map for compound P-3+P-4+P-5 profile, risk assessment, mitigation measures, Privacy Officer sign-off.

Safety gate note: This artifact defines the autonomy boundaries of the system before any code is written. Every Level 3 action identified here must have: (1) a named HITL gate in agentic-logic-spec, (2) a Fallback Protocol that resolves without autonomous action on timeout, and (3) an acceptance criterion in user-stories that tests the false positive scenario. An L3 action without all three is an uncontrolled external commitment — a CRITICAL finding at harness-audit-grader stage.